

Say Less, Never Mislead

Cross-Layer Selective Abstention in an Offline Assistive Perception System, Extended to a Tool-Mediated Voice Agent

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Abstract

Sighted users silently discard a system’s wrong answers; blind users cannot. We report BlindAssist, an offline camera-based guidance system for blind and low-vision users, and argue that *selective abstention*—declining to speak when an output would be unreliable—is a first-class design objective at every layer, not an error-handling detail. We describe five mechanisms spanning perception (reliability-gated metric distance), planning (an openness threshold that can answer “stop”), transport (no-data distinguished from verified-clear), attention (proximity-gated warnings with an on-demand directional counterpart), and dialogue: a tool-mediated voice agent in which a local offline language model selects among fourteen deterministic capabilities and authors no guidance. On 200 labelled utterances, deterministic-first two-tier routing improves overall accuracy from 39.5% to 53.0% while keeping 100% on trained phrasings that an LLM-only router drops to 45.0%, and the same model asked to answer freely rather than to choose a tool fabricates perceptual content in 42.5% of responses. It also costs: out-of-scope over-triggering rises from 5.0% to 55.0%, which we report as a negative result about small local routers rather than tune away.

CCS Concepts

• **Human-centered computing** → **Accessibility technologies**; *Ubiquitous and mobile computing*; • **Computing methodologies** → *Artificial intelligence*.

Keywords

blind and low-vision users, assistive technology, abstention, on-device machine learning, voice agents, tool use, hallucination

1 Introduction

A sighted person using an assistive app performs a silent, continuous audit. The app says “chair on your right”; they glance right; there is no chair; they discard the claim and lose nothing. That audit loop is invisible in design documents precisely because it is free.

For a blind user it does not exist. Every spoken claim is accepted or acted on, because there is no cheap channel to check it against. The asymmetry is not hypothetical: blind readers of machine-generated image captions have been shown to construct elaborate and confident interpretations of descriptions that were simply wrong, because nothing in the interaction offered grounds for doubt [?], and the same pattern recurs with generative assistants, whose fluency is itself read as a reliability signal [?]. The

human-factors literature supplies the other half: unreliable automation does not merely fail to help, it drives disuse of the aid as a whole [?], and below a reliability crossover an imperfect alert is worse than no alert at all [?].

This inverts a default that most interactive systems take for granted: *answer something*. Under the asymmetry, a confidently wrong answer is not a degraded answer—it is worse than silence, because silence costs the user one query and a wrong answer costs them their calibration of when to trust the system at all.

This paper reports BlindAssist, a working offline guidance system (a Python reference implementation and an Android/Flutter port driven by a shared pure-logic core with mirrored test suites), and makes one architectural claim:

Under the verification asymmetry, **selective abstention should be designed per layer, with layer-specific abstention criteria**—not delegated to a single confidence threshold at the output.

Contributions. **C1**, a cross-layer selective-abstention pattern instantiated five times with layer-specific criteria. **C2**, a tool-mediated voice agent for a safety-critical speech channel: the model emits only a validated {tool, argument} pair drawn from a fixed registry, and every *guidance* token originates in deterministic code or a fixed template. **C3**, deterministic-first two-tier routing, which we show is not a latency optimisation with an accuracy cost but strictly better on the traffic it covers. **C4**, a capability registry with enforced cross-site consistency across two languages. **C5**, an evaluation, on a protocol frozen before the router was implemented, in which abstention rate and fabricated-perception count are first-class metrics rather than failure modes.

2 Related work

Assistive perception systems. Crowd- and cloud-backed visual question answering established the interaction [?] and the dataset tradition [?] this system inherits, against a backdrop of everyday interfaces whose visual content is simply unavailable [?]; today’s scene-description apps [?] produce rich descriptions but are latency- and connectivity-dependent, verbose by design, and not built for continuous walking obstacle avoidance. Dedicated wearables [?] give excellent near-field reading at hardware cost. Indoor navigation assistants localise the traveller rather than the objects in front of them [?], spatial-audio wayfinding operates at beacon granularity [?], and electronic travel aids [?] give proximity without object identity or fine direction; surveys [?] and comparative studies [?] map the space. Work on blind users training their own recognisers [?] shares our premise that the recogniser is not the

whole of the problem. What we did not find in this literature is a system that treats *declining to answer* as a designed and evaluated behaviour rather than an error path.

Abstention. Classification with a reject option is old [? ?] and has a modern formal treatment in selective prediction [? ?], learning to defer [? ?], and confidence estimation [? ?]. That work asks when a *model* should abstain, and answers with a score threshold. We treat abstention as an *interface* property instead: five different layers of one system, each with a failure mode of its own, each abstaining on a criterion derived from that failure mode rather than from a shared confidence number.

Hallucination containment. Fluent, unsupported generation is well characterised [? ?], and models’ own confidence signals are only partly informative [?]. Tool use [? ? ? ?] and constrained decoding [? ?] are standard engineering. Our step is not the mechanism but *why* it is applied: for a consumer who cannot visually reject a wrong answer, containment is a safety property, and the design target is not “minimise fabricated perception” but “make it inexpressible”. We report the ablation that prices that constraint.

Non-visual output. Obstacle sonification [? ?] and vibrotactile direction [? ?] are established modalities; our contribution is not the modality but a single ordinal-localisation core that drives speech, stereo sonar and haptics with consistent semantics and shared anti-spam timing.

3 System

A handset streams camera frames to a tethered laptop over Wi-Fi; the laptop runs YOLOv8s [? ?] trained on COCO [?] plus a small custom door/dustbin detector; the handset keeps every output modality native—speech, stereo-panned proximity sonar, haptics, grammar-constrained offline speech recognition [? ?], and on-device OCR. All guidance logic lives in a pure layer with no camera, no model and no clock of its own, mirrored 1:1 between Python and Dart with identical test suites (221 and 159 tests). Figure 1 shows the arrangement.

Objects are localised *ordinally*: a 3×3 zone grid from the box centre, and a per-class proximity bucket (very close / close / medium / far) from box area. Metric distance exists but is gated (§4).

On-device inference measured ≈ 2.5 s per frame on the handset for both the GPU and NNAPI delegates, which is why inference is tethered. On the laptop GPU both detectors together cost 21 ms; total server time per frame during a field walk was ≈ 305 ms, so the dominant cost is frame reconstruction and transport rather than detection.

4 Five ways to decline

Perception: reliability-gated metric distance. Distance comes from a pinhole model, $d = h_{\text{real}} \cdot F / h_{\text{box}}$, and is spoken as “about N metres” only if three gates pass: the box does not touch a frame edge, the class name clears a confidence threshold, and the object is not already close. The edge-clip gate matters most: a truncated box reads *falsely far*, so the error points in the dangerous direction, precisely for the nearest and largest objects. When any gate fails the system speaks the ordinal bucket instead. Learned monocular depth [? ?] would improve the estimate but not remove the need for the gate, since its failure cases are also silent.

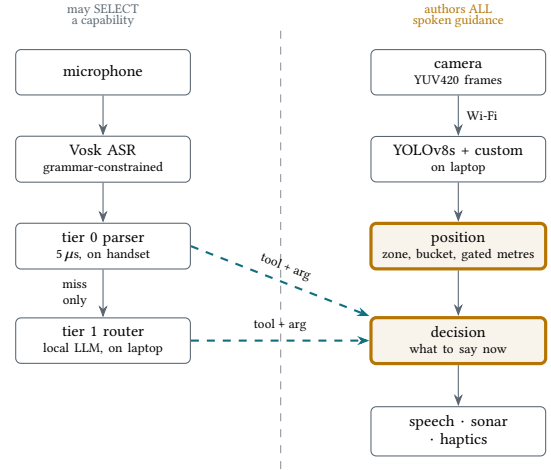


Figure 1: The two dashed arrows crossing the boundary are the whole safety argument: the router may point at a capability, but it may not speak through one. Tier 0 runs on the handset, so every trained phrase still works with the laptop switched off.

Planning: openness-thresholded path advice. Each of left/ahead/right is scored by the proximity rank of its *closest* obstacle rather than by summed occupied area; the naive metric lets a far bulky object outrank a near small hazard and steers the user into it. Doorways are excluded from obstacle mass. If even the emptiest third holds a close obstacle, the system refuses to name a least-bad direction and says “Stop, no clear path”.

Transport: absence is not a negative. A failed frame returns null, never an empty list. An empty list means a *verified-clear* scene, and acting on it silences the sonar, resets escalation state, and makes Find report “not visible” for an object that may be directly ahead. On no-data the engine pauses guidance and says so.

Attention: proximity-gated warnings with a pull counterpart. Added after the first field walk, where the verdict was “it keeps saying all the objects, it’s too much of a cluster”. Continuous warnings now fire only at close range or nearer; everything removed from the push channel is reachable by asking (check: “is there anything in front of me?”), answered by the same ordinal core, in tier 0, with no model and no network. The reasoning is the alarm-fatigue result [? ?] applied to a user whose only guidance channel is the one being flooded: an unheeded warning has negative value, because it spent attention and delivered nothing.

Dialogue: routing abstention. Unknown tool, unknown object class, missing argument, prose instead of JSON, timeout or any exception all become abstention, and the clarifying question is selected from a fixed table by key rather than written by the model.

5 The agent layer

BlindAssist’s offline recogniser is grammar-constrained: only phrases in an explicit list can be transcribed *at all*, so free speech is not mis-parsed—it is never heard. A trigger word therefore opens a dictation window, transcribed either by a local Whisper model [?] on the laptop or, with the laptop off, by a second recogniser built on

the same already-loaded 40 MB model with its grammar removed. The transcript goes first to the deterministic parser and only then, on a miss, to the router (Figure 2).

The authority boundary. The model receives the tool registry, the class enumeration, a deterministic state block (visible classes with zone, proximity and count; mode; last announcement; object memory) and the utterance, and must return JSON. Everything after that treats its output as untrusted input. Multi-turn references such as “is it still there” therefore resolve against the detector’s own output, never against a description: perception never re-enters the model.

Where the boundary moved. The system as first built enforced the stronger rule—*no spoken token whatsoever* originated in the model. After the first field walk the developer-user asked for conversation: the ability to ask a question in their own words and be answered rather than routed. We granted exactly that. A reply travels in a separate channel that the executor cannot route through any capability; it is grounded in the same state block, length-capped and truncated at a sentence; loose prose emitted where a tool call belongs is still discarded; and one flag restores the absolute rule for the ablation. The honest formulation of C2 is therefore “no *guidance* token originates in the model”, which is weaker than where we started, and we report it as weakened.

Tiering across a network boundary. On the handset the two tiers fall on opposite sides of a Wi-Fi link. The client re-validates the server’s reply against the same closed registry before executing, so the containment guarantee does not rest on trusting the transport, and a reply containing one unusable action is discarded whole—executing the half that happened to parse is itself an unverified action.

Registry. Fourteen capabilities (Table ??) were previously declared in four places: a parser, its phrase list, a dispatcher, and the Dart equivalents. The sites had already drifted—the web dispatcher silently dropped five capabilities the parser could produce. One declarative table now drives the recogniser’s phrase list at runtime, generates the model’s tool schema, backs a single executor, and emits a committed manifest that both languages’ test suites assert against field by field. We claim *enforced consistency across sites*, not literal single-source generation: the field-validated parser is deliberately left in place rather than regenerated, because the regression risk of rewriting it exceeds the value of the stronger claim.

6 Evaluation

The perception and guidance layers were checked first on seven recorded phone clips, which produced 31 announcements; every announcement saves an annotated keyframe, and all 31 were reviewed against the image. Direction was correct on 31/31 and there were no phantom announcements, while 6 of 31 carried a wrong class name with the warning behaviour still correct—the pattern that motivates speaking the generic word “obstacle” below a confidence threshold. These numbers are small and author-reviewed, and we report them as a sanity check on the deterministic core rather than as a perception result. The rest of this section concerns the dialogue layer.

The protocol and a set of 200 labelled utterances were frozen and committed *before* the router was implemented, so the router

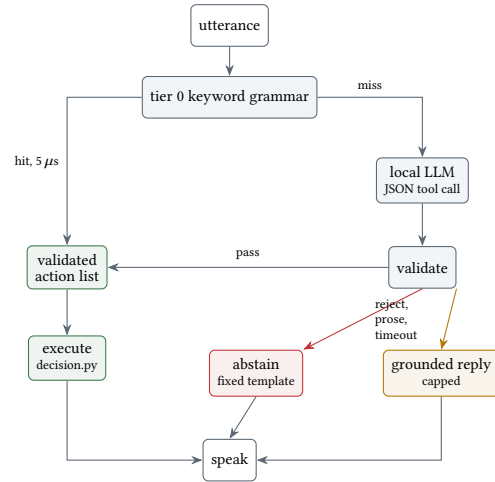


Figure 2: Two-tier routing. Every path except the amber reply channel produces text written by deterministic code or selected from a fixed template; the reply channel cannot reach any capability’s output.

could not be tuned against its own test set. Categories: *canonical* (40, phrasings the grammar covers), *paraphrase* (70), *multi-intent* (20), *out-of-scope* (40, gold label: abstain) and *ambiguous* (30, gold depends on a state block encoded in the record). Out-of-scope records are plausible things a user would say to an assistive device—weather, calls, messages, time, battery—rather than nonsense strings, which would be easy to abstain on and would inflate the metric.

Configurations: keyword-only, LLM-only (tier 0 stubbed to always miss), and two-tier. Routing accuracy is exact match on the *ordered* action list; a two-action utterance routed to one correct action scores zero, because the user asked for two things and got one. The *over-trigger rate* is the fraction of out-of-scope utterances answered with any action, and is the safety metric of this layer. The fabrication check verifies that every executed spoken string is a member of the set the deterministic core could produce for that record, plus the fixed templates; the harness fails loudly otherwise. Intervals are Wilson [?]; no significance testing between configurations is claimed at this set size, and the comparison is descriptive.

Model: 11ama3.2:3b [?] under Ollama [?] on the tether laptop. Criteria that would falsify each claim were recorded in the protocol before the runs.

7 Results

Three things in Table 1 matter more than the overall column.

The baseline’s shape is the motivation stated as data. It is perfect on the phrasings it was designed for and *zero* on paraphrase and multi-intent—not degraded, absent.

Tiering strictly dominates LLM-only. LLM-only loses 55% of the canonical category: the model mis-routes utterances the keyword parser gets right by construction, usually by substituting a plausible neighbour (describe for count, walk for zones). Two-tier keeps 100% there because those utterances never reach the

Table 1: Routing accuracy (%). Wilson 95% CIs on the overall row: keyword 33.0–46.4, LLM-only 38.3–51.9, two-tier 46.1–59.8.

Category	<i>n</i>	keyword	LLM-only	two-tier
canonical	40	100.0	45.0	100.0
paraphrase	70	0.0	48.6	47.1
multi-intent	20	0.0	30.0	10.0
out-of-scope	40	95.0	47.5	45.0
ambiguous	30	3.3	43.3	43.3
overall	200	39.5	45.0	53.0

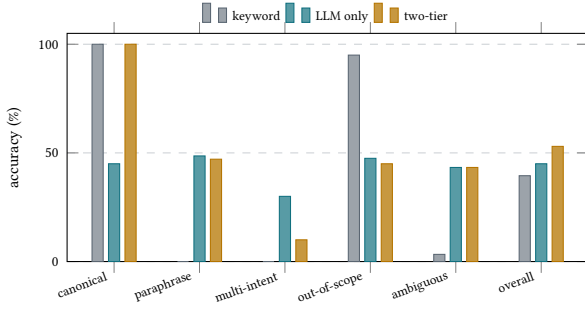


Figure 3: Routing accuracy by category, $n = 200$. For out-of-scope, “accuracy” means correctly abstaining. The keyword bars at zero are not degradation but absence: a grammar cannot hear what is not in its grammar.

model, and still collects most of the paraphrase gain. The deterministic tier is not a latency optimisation with an accuracy cost; it is more accurate *and* faster on the traffic it covers.

The abstention collapse is the headline, and it is negative. The keyword baseline abstains on 95% of out-of-scope input for free—an unmatched utterance returns nothing. Both language-model configurations spend nearly all of it (Table 2): a 3B model choosing among fourteen tools nearly always finds one it likes. “Call my mum” becomes read; “what time is it” becomes clock; “take a photo” becomes walk. We report this unmitigated because the protocol was frozen before the router existed and tuning the prompt against this set is what the freeze forbids. It does not falsify C1 or C2—no run fabricated perception, and tier-0 abstention is intact for the traffic tier 0 covers—but it falsifies any reading of C5 in which adding a local model is a free improvement. Coverage and abstention trade against each other at this model size, and the two-tier structure is what keeps the trade from reaching trained phrasings.

Both keyword over-triggers are substring collisions: “read my email” contains *read*; “how do i get to the bus stop” contains *stop*. They are the price of matching on keywords, and they are exactly the errors a router with sentence-level context should remove—it does not, because tier 0 claims them before the model is consulted. That is the cost side of C3 stated plainly.

Per-call tier-1 latency is essentially identical across the two configurations (1172 vs. 1188 ms at p50), so two-tier’s latency advantage

Table 2: Out-of-scope behaviour ($n = 40$) and routing latency. Wilson 95% CIs on the over-trigger rates: keyword 1.4–16.5, LLM-only 37.5–67.1, two-tier 39.8–69.3.

	keyword	LLM-only	two-tier
abstained	38	19	18
answered anyway	2	21	22
over-trigger rate	5.0%	52.5%	55.0%
tier-0 hit, p50	5 μs	—	5 μs
tier-1 hit, p50	—	1172 ms	1188 ms
tier-1 hit, p95	—	1578 ms	1500 ms
traffic served by tier 0	30.0%	0%	30.0%
conversational replies	0	4	6

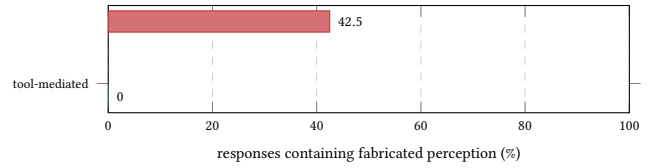


Figure 4: The same model, the same deterministic state block. Asked to *choose a tool*, it fabricated nothing in any configuration (0 of 200); asked to *answer*, it fabricated in 42.5% of responses (85 of 200). The zero is by construction, not by tuning; the 42.5% is a keyword-based lower bound counting invented objects only.

is entirely a question of *how much* traffic reaches the model: 30% of utterances are served at 5 μ s, and the trained commands users issue most often are exactly the 30%. Tier 1 is usable for on-demand questions and unusable inside a continuous guidance loop; every capability here is on-demand, which is a fact about this system rather than a general result.

Six two-tier responses and four LLM-only ones were conversational replies rather than actions (§5). A reply carries no action, so it scores as an abstention and is not an over-trigger; we list the counts so the reader can subtract them.

Figure 4 is the clearest result in the paper, and the character of the inventions is worse than the rate suggests. Asked to “find bottle” with a state block listing no bottle, the model replied:

“i’m walking in front of you, my cane tapping on the ground. i’ve stopped about 6 feet away from your right side. there’s a small...”

It invents an object, a distance in feet, a bearing, and a first-person travelling companion with a cane. For a sighted user this is a curiosity to dismiss. For the user this system is built for it arrives in the same voice, at the same volume, with the same confidence as a real detection.

A capability added after the freeze. The check capability was implemented after the set was committed. Two records are affected and neither was re-labelled: one paraphrase whose gold we consider superseded, and one ambiguous record whose failure mode changed from silent abstention to a confident answer to a different question—strictly worse, and the predictable cost of widening a

keyword grammar. A third instance was caught by the frozen out-of-scope category before any user met it: the first implementation treated any “left” as a direction, so “how much battery is left” routed to a scene query, raising over-trigger to 7.5%. “Left” and “right” are ordinary English words; “ahead”, “front” and “forward” are not, so the rule now requires a positional lead-in for the ambiguous pair only.

8 Discussion and limitations

No blind participants. The system has been field-walked by a single sighted developer. Every claim here about what is *safer* is a design argument supported by mechanism and by deterministic tests, not by user data. A study with blind participants is the necessary next step and specifically the one that could falsify the premise: users may prefer a guessed answer to an abstention, and the over-trigger rate we optimise against may not be the quantity that matters to them.

Over-triggering is measured, not solved. 55% is the largest open problem this evaluation exposes. Rejection examples in the prompt, a second-pass scope classifier, and a model-confidence gate [?] are all available; none is evaluated, because each would be tuning against a frozen set. A held-out set comes first.

Clean text, not speech. Routing is evaluated on written utterances, so recognition errors are outside the loop. The protocol specifies a spoken-transcript condition with speakers who did not author the set; it is not reported here, and paraphrase and out-of-scope numbers in particular should be read as upper bounds on what the deployed pipeline achieves.

One model, one machine. A second arm with a reasoning model (qwen3:4b) returned no parseable tool call on 102 of 140 tier-1 calls at 6–8 s each, so tier 1 contributed nothing and the run scored exactly the keyword baseline. The validation boundary degraded to fewer capabilities rather than to wrong ones, which is the designed behaviour, but the comparison is not informative about model size until thinking output is disabled.

The clock mapping is camera-frame, not O&M. Frame width spans 10–2 o'clock over roughly a 60° field of view, so “2 o'clock” means the right frame edge, not the 90° a trained traveller would turn. This needs relabelling or a genuine remapping before the system claims compatibility with Orientation & Mobility instruction.

Distance is coarse and the fabrication detector is crude. Roughly ± 30 –40% at 5 m with an uncalibrated focal constant; the gating argument concerns *when* to speak a number, not how good it is, but a one-time per-device calibration would strengthen any accuracy claim. The fabrication detector flags invented objects only, missing invented distances, bearings and counts—several of which appear in the samples it did flag. It is a lower bound.

Chat replies are counted, not judged. They are excluded from the fabrication metric by definition and listed for inspection, but we do not measure whether they are correct or well-calibrated. They are now part of what the system says, so a user study would have to.

Tether dependency. The remote-primary architecture assumes a laptop and a local network; §4 makes the failure safe, not absent. On-device viability awaits a lighter detector head or hardware whose delegate partitions the model cleanly.

9 Ethics, positionality and availability

The author is a sighted student developer and is not a member of the population this system is built for. Nothing here was co-designed with blind users, and the one field walk was performed by the author. We have tried to make that limitation load-bearing rather than decorative: every safety claim is tied to a mechanism and a test that a reader can inspect, so that a blind participant study can falsify the design arguments rather than merely fail to confirm them. No human-subjects data was collected and no ethics approval was required for the work reported; a participant study is not attempted here precisely because doing it informally, with an unvalidated prototype and no approval, would be the wrong way to involve the population concerned.

The system runs offline by design—detection on a locally tethered laptop, speech recognition, synthesis and OCR on the handset—so no camera frame, no utterance and no location leaves the user’s own devices. The tether is an unencrypted local link, which is adequate for a prototype on a personal hotspot and would need transport security before any deployment.

Source code, the capability registry manifest, the frozen evaluation protocol, the 200-record labelled set and every run report are available at https://github.com/Aditya17-bot/object_detection_blind. Results in this paper carry the eval-set SHA-256 prefix e4eeca83070e2d66; any number quoted from a set with a different hash is not comparable.

10 Conclusion

For users who cannot audit what a system tells them, abstention is not an error path—it is a feature that must be designed, implemented and measured at every layer where the system can be wrong. We showed five such designs in a working offline assistive system, each with criteria specific to its layer’s failure mode, and extended the pattern to a voice agent whose language model may choose what the system does and never what it says about the world. The interesting claim is not that tool mediation prevents fabrication—it plainly does—but that the same principle that motivates it also explains a distance gate, a path threshold, a null-versus-empty distinction and a warning-rate ceiling three layers away.

References

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